



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AI Vibe Coding for Python Applications

Course Code: C179

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	22 July 2026	Initial release of course C179 — 3 days, 5 topics and 20 hands-on labs building the CardGuard fraud-screening application with an AI coding assistant. Day 3 closes with an end-to-end consolidation build and an extended course-wide recap.	Dr. Alfred Ang

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Course Information

Course Title	AI Vibe Coding for Python Applications
Course Reference	C179
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	3 days · 7.5 instructional hours per day (22.5 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on practical labs
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply vibe coding practices — set up a reproducible Python project with uv, and use an AI coding assistant to scaffold, explain and refactor working code.
- LO2: Design and implement object-oriented Python — classes, encapsulation, inheritance and composition — using AI-assisted conversational design.
- LO3: Build data analytics pipelines with pandas — load, clean, transform, group and aggregate realistic datasets into reportable results.
- LO4: Model and validate data with Pydantic and expose it through a FastAPI application with typed request and response models.
- LO5: Package and deploy a Python application — dependency locking, configuration, containerisation and a running deployed service.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Vibe Coding Foundations & Object-Oriented Python

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and ice-breaker
10:00–10:45	45 min	Topic 1 — Vibe Coding Foundations: uv Projects · AI Coding Assistants · Prompting Patterns · AI-Assisted Refactoring (concepts + demo)
10:45–11:00	15 min	Tea break

11:00–13:00	120 min	Hands-on: Lab 1: Set Up a Reproducible Python Project with uv; Lab 2: Write Effective Prompts for an AI Coding Assistant; Lab 3: Review and Correct AI-Generated Code
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Hands-on: Lab 4: Refactor Working Code Conversationally. Topic 2 — Object-Oriented Programming in Python: Classes · Encapsulation · Inheritance · Composition · Dunder Methods (concepts)
15:30–15:45	15 min	Tea break
15:45–18:15	150 min	Hands-on: Lab 5: Model a Domain with Classes and Dunder Methods; Lab 6: Encapsulate State with Properties and Validation; Lab 7: Build a Rule Hierarchy with Inheritance and Polymorphism; Lab 8: Compose Rules into an Engine
18:15–18:30	15 min	Day 1 recap and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 2 — Data Analytics with pandas, Pydantic & FastAPI

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and Q&A
9:45–10:45	60 min	Topic 3 — Data Analytics with pandas: DataFrames · Cleaning · Transformation · GroupBy · Aggregation · Reporting (concepts + demo)
10:45–11:00	15 min	Tea break
11:00–13:00	120 min	Hands-on: Lab 9: Load and Explore Transactions with pandas; Lab 10: Handle Errors with Exceptions
13:00–14:00	60 min	Lunch break
14:00–15:30	90 min	Hands-on: Lab 11: Build Per-Cardholder Baselines with GroupBy; Lab 12: Detect Velocity Bursts with Rolling Time Windows
15:30–15:45	15 min	Tea break
15:45–18:15	150 min	Topic 4 — Data Modelling with Pydantic and FastAPI: Typed Models · Validation · Endpoints · Request/Response Schemas · Auto Docs (concepts). Hands-on: Lab 13: Replace Hand-Written Validation with Pydantic Models; Lab 14: Expose the

		Screening Engine as a FastAPI Endpoint
18:15–18:30	15 min	Day 2 recap and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 3 — FastAPI, Packaging, Deployment & Consolidation

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and Q&A
9:45–11:00	75 min	Hands-on: Lab 15: Persist Screening Decisions to SQLite; Lab 16: Test the API with pytest and httpx
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Topic 5 — Packaging and Deployment: uv Lockfiles · Configuration · Containers · Deploying a Service (concepts). Hands-on: Lab 17: Build a Fraud Analyst Dashboard with Streamlit
13:00–14:00	60 min	Lunch break
14:00–15:45	105 min	Hands-on: Lab 18: Externalise Configuration and Protect Secrets; Lab 19: Lock Dependencies and Containerise the API; Lab 20: Deploy the Full Stack with Docker Compose
15:45–16:00	15 min	Tea break
16:00–17:30	90 min	Consolidation build: rebuild the CardGuard stack end-to-end from a clean checkout — uv sync, run the analytics pipeline, start the FastAPI service and the Streamlit dashboard, then containerise and verify the health endpoint. Trainer-supported, working individually.
17:30–18:30	60 min	Course-wide recap and Q&A: the four-part prompt pattern, the OOP/analytics/API/deployment spine, common AI-generated defects and how each was caught, and a guided walkthrough of where to take CardGuard next.

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Vibe Coding Foundations	—	Lab 1, Lab 2, Lab 3, Lab 4

Topic 02: Object-Oriented Programming in Python	—	Lab 5, Lab 6, Lab 7, Lab 8
Topic 03: Data Analytics with pandas	—	Lab 9, Lab 10, Lab 11, Lab 12
Topic 04: Data Modelling with Pydantic and FastAPI	—	Lab 13, Lab 14, Lab 15, Lab 16
Topic 05: Packaging and Deployment	—	Lab 17, Lab 18, Lab 19, Lab 20